Creacio d’una imatge desde zero



Óscar Benítez i Esteve Amargant
Josep Escoda
Optativa Desplegament k8s

Practica 1

Codi font a hw.c

```
➜ practica1 git:(development) X cat hw.c
#include <stdio.h>

int main(){
        printf("Hooola mon!\n");
        return 0;
}
➜ practica1 git:(development) X
```

El compilem i comprovem que funcioni

```
}
➜ practica1 git:(development) X gcc -static -o hw hw.c
➜ practica1 git:(development) X chmod +x hw
➜ practica1 git:(development) X ./hw
Hooola mon!
➜ practica1 git:(development) X
```

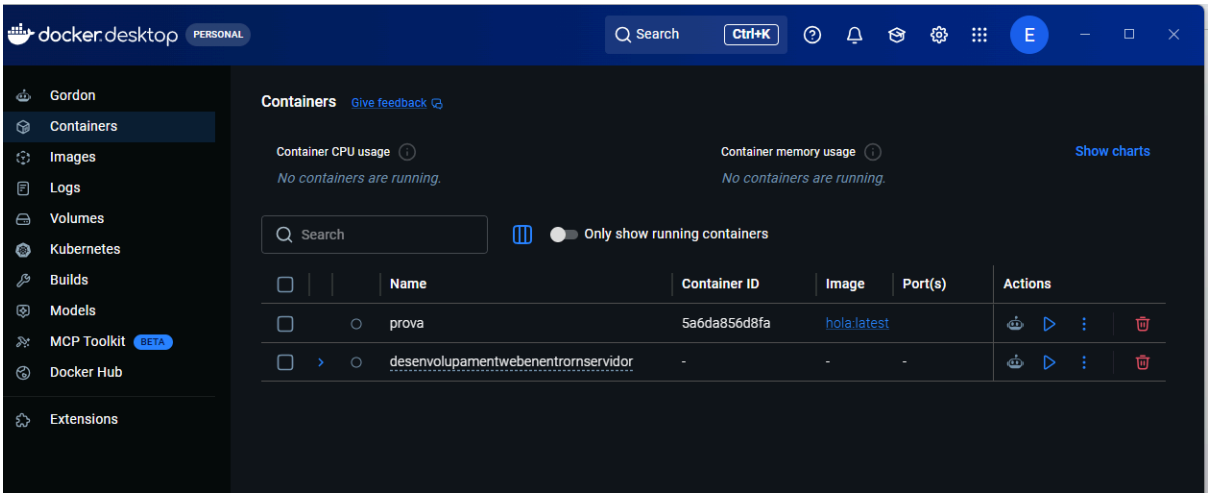
Dockerfile

```
➜ practica1 git:(development) X cat Dockerfile
FROM scratch
COPY hw /
CMD ["/hw"]
➜ practica1 git:(development) X
```

Creem la imatge i arranquem el contenidor:

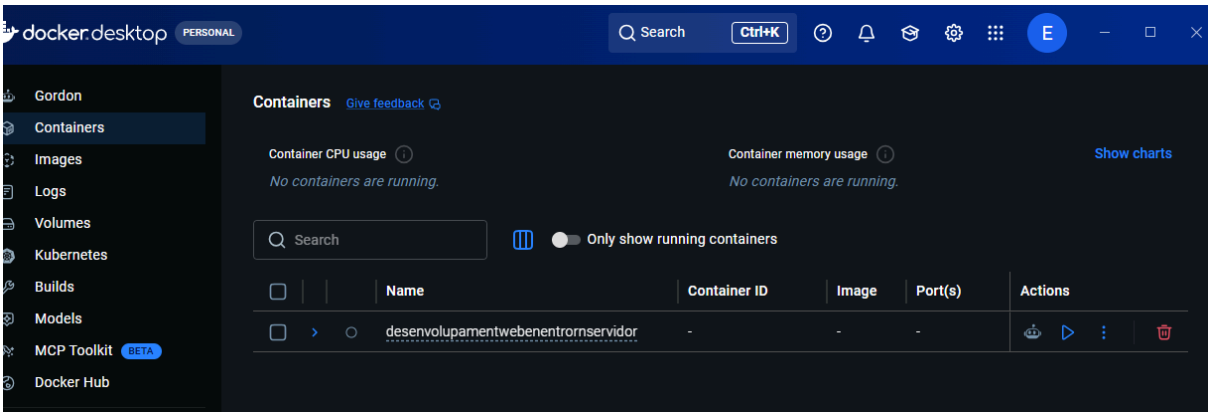
```
➜ practica1 git:(development) X docker build -t hola:latest .
[+] Building 0.6s (5/5) FINISHED                                                                                    docker:default
 => [internal] load build definition from Dockerfile                                                                          0.0s
 => => transferring dockerfile: 72B                                                                                           0.0s
 => [internal] load .dockerignore                                                                                             0.0s
 => => transferring context: 2B                                                                                               0.0s
 => [internal] load build context                                                                                             0.0s
 => => transferring context: 817.11kB                                                                                         0.0s
 => [1/1] COPY hw /                                                                                                           0.0s
 => exporting to image                                                                                                        0.3s
 => => exporting layers                                                                                                       0.1s
 => => exporting manifest sha256:a8bc5794ba59897e29231efb2f9095051f2d82aef8582046bf126f3650a040f7                             0.0s
 => => exporting config sha256:6cfc61fb126ee2dfacf5777b6a9407de87ec902aed36bd0275552e4f69f3144e                               0.0s
 => => exporting attestation manifest sha256:6521606db2c46b880610f6d01101669efdefc03469872fd88a28ef21230c5728                 0.0s
 => => exporting manifest list sha256:398d850fb99427ebae6f4c7350576ebc33e88b9cf8b4a818b774ff1aac89ec52                        0.0s
 => => naming to docker.io/library/hola:latest                                                                                0.0s
 => => unpacking to docker.io/library/hola:latest                                                                             0.0s
➜ practica1 git:(development) X docker run --name prova hola:latest
Hooola mon!
➜ practica1 git:(development) X
```

Name	Container ID	Image	Port(s)	Actions
prova	5a6da856d8fa	hola:latest		



I eliminem el contenidor i la imatge

```
→ practica1 git:(development) X docker run --name prova hola:latest
Hooola mon!
→ practica1 git:(development) X docker rm prova
prova
→ practica1 git:(development) X docker rmi -f hola:latest
Untagged: hola:latest
Deleted: sha256:398d850fb99427ebae6f4c7350576ebc33e88b9cf8b4a818b774ff1aac89ec52
→ practica1 git:(development) X
```



gcc -static -o hw hw.c: gcc és el compilador de C, -static enllaça les llibreries.-o hw indica el nom del fitxer de sortida i hw.c és el fitxer amb el codi font.
chmod +x hw: Canvia els permisos del fitxer. +x afegeix permís d'execució ./hw Executa el binari del directori actual.
docker build -t hola:latest: Construeix una imatge a partir del Dockerfile. -t posa nom i etiqueta a la imatge. `.` és el directori on hi ha el Dockerfile i els fitxers a copiar.
docker run --name prova hola:latest: Crea i arrenca un contenidor a partir de la imatge. --name prova dona nom al contenidor.
docker stop prova: Atura el contenidor prova.
docker rm prova: Elimina el contenidor.
docker rmi -f hola:latest: Elimina la imatge. -f força l'eliminació.

Practica 2

app.py

```
→ practica2 git:(development) X cat app.py
import os
import socket
from flask import Flask
from redis import Redis, RedisError

# Connect to Redis
redis = Redis(
    host=os.getenv("REDIS_HOST", "localhost"),
    port=6379,
    db=0,
    socket_connect_timeout=2,
    socket_timeout=2,
)

app = Flask(__name__)


@app.route("/")
def hello():
    try:
        visits = redis.incr("counter")
    except RedisError:
        visits = "<i>cannot connect to Redis, counter disabled</i>"

    html = (
        "<h3>Hoooola {name}!</h3>"
        "<b>Hostname:</b> {hostname}<br/>"
        "<b>Visits:</b> {visits}"
    )
    return html.format(
        name=os.getenv("NAME", "world"),
        hostname=socket.gethostname(),
        visits=visits,
    )


if __name__ == "__main__":
    app.run(host="0.0.0.0", port=80)
→ practica2 git:(development) X
```

REqueriments.txt

```
→ practica2 git:(development) X cat requirements.txt
Flask
Redis
→ practica2 git:(development) X
```

Docekerfile

```
→ practica2 git:(development) X cat requirements.txt
Flask
Redis
→ practica2 git:(development) X cat Dockerfile
FROM python:3.10-slim

RUN apt-get update && apt-get install -y \
    redis-server \
    && rm -rf /var/lib/apt/lists/*

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY app.py .

EXPOSE 80

CMD ["sh", "-c", "redis-server --daemonize yes && python app.py"]
→ practica2 git:(development) X
```

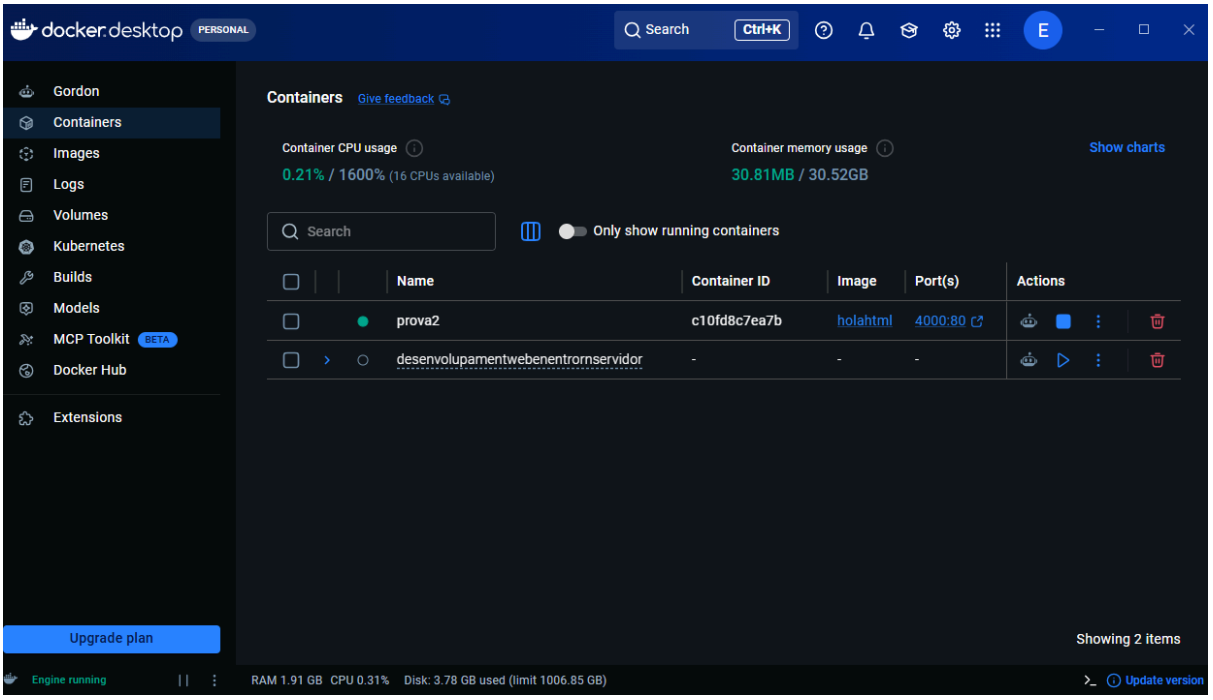
Creem la imatge:

```
→ practica2 git:(development) X docker build -t holahtml .
[+] Building 2.0s (12/12) FINISHED                                                                                  docker:default
 => [internal] load build definition from Dockerfile                                                                          0.0s
 => => transferring dockerfile: 342B                                                                                          0.0s
 => [internal] load metadata for docker.io/library/python:3.10-slim                                                           1.2s
 => [auth] library/python:pull token for registry-1.docker.io                                                                 0.0s
 => [internal] load .dockerignore                                                                                             0.0s
 => => transferring context: 2B                                                                                               0.0s
 => [1/6] FROM docker.io/library/python:3.10-slim@sha256:6ff506466f8b1e981719b468cc0023b8ef000b983cdec0eafe119d01             0.0s
 => => resolve docker.io/library/python:3.10-slim@sha256:6ff506466f8b1e981719b468cc0023b8ef000b983cdec0eafe119d01             0.0s
 => [internal] load build context                                                                                             0.0s
 => => transferring context: 63B                                                                                              0.0s
 => CACHED [2/6] RUN apt-get update && apt-get install -y     redis-server     && rm -rf /var/lib/apt/lists/*                  0.0s
 => CACHED [3/6] WORKDIR /app                                                                                                 0.0s
 => CACHED [4/6] COPY requirements.txt .                                                                                      0.0s
 => CACHED [5/6] RUN pip install --no-cache-dir -r requirements.txt                                                           0.0s
 => CACHED [6/6] COPY app.py .                                                                                                0.0s
 => exporting to image                                                                                                        0.5s
 => => exporting layers                                                                                                       0.0s
 => => exporting manifest sha256:7b5a8cdbbbac5117a40b52674047378372c2342314759c306f252bfdb49dfa99                             0.0s
 => => exporting config sha256:beff49be9564f05d119472e5ce4b49ce38d537cb74704d5af6cf42ea123530a2                               0.0s
 => => exporting attestation manifest sha256:b3b07813629be94438d12321a7b203a80e2ce373b218592880eed6ab39e3695d                 0.0s
 => => exporting manifest list sha256:0737ac44bb951b8528897c23ec80fe7843a332f924c8576198c95e8b76b99669                        0.0s
 => => naming to docker.io/library/holahtml:latest                                                                            0.0s
 => => unpacking to docker.io/library/holahtml:latest                                                                         0.4s
→ practica2 git:(development) X
```

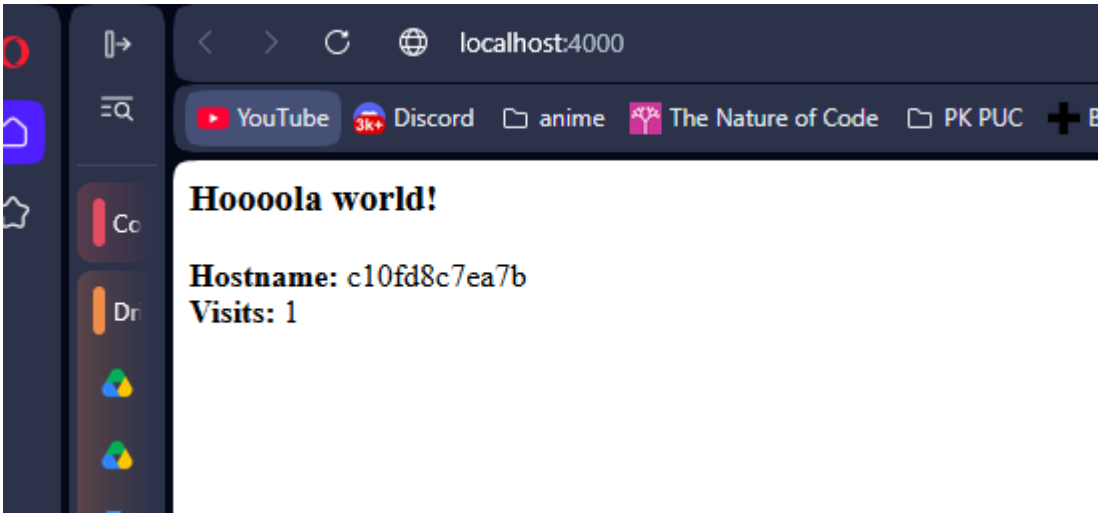
i arranquem el contenidor

```
→ practica2 git:(development) X docker run --rm --name prova2 -p 4000:80 holahtml
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:80
 * Running on http://172.17.0.2:80
Press CTRL+C to quit
```

	Name	Container ID	Image	Port(s)	Actions
☐	● prova2	c10fd8c7ea7b	holahtml	4000:80	
☐	› ○ desenvolupamentwebenentrornservidor	-	-	-	



I entrem a localhost



FROM python:3.10-slim: Imatge base → Debian amb Python 3.10 ja instal·lat.
RUN apt-get update && apt-get install -y redis-server &&
rm -rf /var/lib/apt/lists/*: Actualitza l'índex de paquets, instal·la el servidor Redis (-y respon "sí" a tot).
WORKDIR /app: Estableix i crea /app com a directori de treball per al contenidor.
COPY requirements.txt: Copia el fitxer de dependències a /app.
RUN pip install --no-cache-dir -r requirements.txt: Instal·la Flask i Redis. -r llegeix la llista del fitxer, --no-cache-dir no guarda pip.
COPY app.py: Copia el codi de l'aplicació a `/app`.
EXPOSE 8`: El contenidor escolta al port 80.
CMD ["sh", "-c", "redis-server --daemonize yes && python app.py"]: Arrenca Redis en segon pla , --daemonize yes i si va bé && arrenca l'app Flask.